BarterSwap - Student E-Commerce Platform Project Proposal

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1. Problem Statement

Many students own goods such as books, electronics, or clothing that they no longer use. However, they do not have an efficient and secure platform to trade these items within their community. Existing solutions are often paid, publicly available, or unsafe for students, and they usually rely on real money transactions.

2. Vision

BarterSwap is a web-based application designed to serve as a peer-to-peer marketplace exclusively for students. It allows them to buy, sell, and exchange goods using a virtual currency system within a secure, student-only environment. The platform combines functionalities of e-commerce, bidding systems, and reputation management to create a sustainable student trading ecosystem.

3. Scope

The platform will include user registration, item listings, virtual currency-based bidding, a feedback system, and private messaging. It will also feature dashboards for users and admins to view advanced statistical reports.

4. Risks

- Security risks in storing and handling user data.

- Users not completing transactions after bidding.

- Managing virtual currency fairness and preventing exploits.

- Ensuring consistent UX across devices and browsers.

5. Stakeholders

Name	Description	Responsibilities
End Users (Students)	University students who will use the BarterSwap platform	List items, place bids, perform transactions using virtual currency, provide feedback, and contribute to user experience.
Project Manager	Technical lead or team leader responsible for project delivery	Plans and tracks project progress, manages timeline, assigns tasks, and communicates with stakeholders.
Developer	Developers working with React, Node.js, and PostgreSQL	Implement and maintain features, ensure functional requirements are met, write and test code, and fix bugs.
Database Administrator (DBA)	Person responsible for database design and management	Implements data models, optimizes database performance, ensures data integrity and security.
Tester	Team or individual in charge of software quality assurance	Ensures software works as intended, conducts user scenario testing, identifies and reports bugs and issues.

6. Technologies to Be Used

Frontend	React.js
Backend	Node.js + Express.js
Database	PostgreSQL
ORM	Sequelize
Dev Tools	Visual Studio Code, Insomnia
DB Viewer	JetBrains DataGrip
Version Control	GitHub

7. Major Features

User Features

- Register & Login system (with hashed passwords)
- View & edit personal profile
- Create, update, deactivate listings
- Place bids using virtual currency
- Private messaging system
- View user reputation & feedback
- Dashboard: user-specific statistics (items sold, bids, earnings, spending)
- User can follow other users

Admin Features

- View all users, items, transactions
- Generate complex statistical reports
- View reputation history and activity logs

8. Bidding System Overview

Each item listing can be configured as a fixed price sale or an auction-style listing. Users place bids using their virtual currency. The highest bidder at the end of the auction wins the item, and the transaction is recorded.

9. Security Considerations

- All passwords will be hashed using bcrypt
- JWT-based authentication for secure access
- Input validation and SQL injection prevention
- Role-based access control (admin, user)
- HTTPS and secure cookie storage
- Periodic database backups

10. Dashboard & Reporting Section

A dynamic dashboard will provide data-driven insights to both users and admins. At least three complex reports will be included:

Admin Reports

- Top 5 Most Bid Items
- Top Sellers by Transaction Count
- Monthly Transaction Volume

User Dashboard

- Personal sales and purchases summary
- Total earned vs spent virtual currency
- Bid engagement on user's items
- Item listing success rate (% sold)

11. Feature Release Plan

Need	Priority	Features	Planned Release
Basic system access and identity	High	User registration login/logout	Release 1
Item listing functionality	High	Allow users to create, edit, and deactivate item listings with basic item information	Release 1
Browsing and searching	High	View all active listings, filter by category, and search items using keywords	Release 1
Bidding mechanism	High	Enable users to place virtual currency bids on active items and track the highest bidder	Release 2
Virtual currency system	High	Users start with initial balance and use it for transactions; track balance per user	Release 2

Messaging system	Medium	Allow users to send private messages to communicate about listings	Release 2
Transaction recording	High	Record and store completed transactions between buyers and sellers	Release 2
Reputation and feedback	Medium	Enable buyers to rate sellers and display reputation on profile	Release 3
Admin dashboard	Medium	Admin can view overall platform stats, user activity, and flagged content	Release 3
Reporting and analytics	Medium	Generate insights such as most traded items, active users, and auction success rates	Release 3
Advanced filtering/search	Low	Add filters like price range, condition, and sort by newest/highest bid	Release 3
Task-based currency rewards	Low	Users can earn virtual currency by writing feedbacks	Release 3

11. Mock Data Generation

The PostgreSQL database will be populated using randomly generated seed data for development and testing purposes. Faker.js will be used to simulate realistic usernames, descriptions, prices, etc.

12. Future Enhancements

- Notification system for new bids/messages
- Enhanced filtering and sorting in marketplace
- Gamification (earn currency via tasks, missions)

13. Conclusion

BarterSwap creates a unique, student-exclusive trading ecosystem. It promotes sustainable use of goods, community interaction, and a fun virtual currency model. This platform offers students a practical and secure way to engage in commerce, learn digital trading mechanics, and manage virtual currency in a controlled environment.